

CW
Gnu Linux

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GNU's advantage:

- GNU is a free software project aiming to create a completely free operating system without reliance on proprietary software.
- Free software means the freedom to run, study, modify, and share the software, opposing restrictive proprietary licenses.
- The GNU General Public License (GPL) was developed to ensure software freedom for users and developers.

Kernal:

Kernal generally

The kernel is the core of an operating system that manages communication between software and hardware. Its main tasks include managing hardware resources, processes, memory, and devices.

Kernal module

Kernel modules are extensions that enhance the functionality of the kernel without requiring a full rebuild. They are used to add device drivers or specialized features.

Systemd:

- Systemd is an init system responsible for managing the system boot process and services. It handles background processes, daemons, and system stability.
- Some criticize Systemd for violating the Unix philosophy of creating small, modular tools for specific tasks, arguing that it adds unnecessary complexity to the system.
- Systemd services are processes running in the background to handle tasks like network management, task scheduling, and system functionalities.

Graphicx:

Display server protocol

A display server protocol is an interface between the operating system and the graphics hardware to render visuals on the screen. Examples include X11 and Wayland.

Desktop environment or windows man-ager

A window manager controls the appearance and behavior of windows, while a desktop environment is a complete user interface that includes a window manager, menus, tools, and settings for a full experience.

Some examples

Window managers: i3, Openbox, Fluxbox Desktop environments: GNOME, KDE Plasma, XFCE

Ricing:

Meaning of ricing

Ricing refers to customizing the graphical appearance of a Linux system to make it visually appealing. It involves tweaking themes, icons, window manager configurations.

Favorite ricing

- Openbox
- XFCE

Suckless software:

Explane of Suckless software

Suckless Software is a collection of lightweight and minimalist tools designed for simplicity, speed, and usability by advanced users.

Philosophy of software

Their philosophy revolves around minimalism, adhering to the principle of "less is more" to make code readable, simple, and user-modifiable.

Shells:

POSIX-compliant shell

A POSIX-compliant shell, like Bash or Zsh, adheres to POSIX standards for ensuring compatibility and consistent behavior across various operating systems.

Use full command:

(A):

```
grep -r "hello" testDir -- wc -l
```

(B):

```
ls -l allthefiles.txt
```

(C):

```
find testDir -type f -name "*.c" -- wc -l
```